

Customer Shopping Behavior Analysis

A end-to-end data project spanning cleaning, modeling, SQL analysis, and business insight — built on 3,900+ real customer records.

PYTHON · SQL · SQLITE · PANDAS



Made with GAMMA

Project at a Glance

3,900+

Customer Records

Raw entries cleaned and prepared for analysis

4

Core Phases

Cleaning, Modeling, EDA, and Business Insights

5+

SQL Techniques

CTEs, aggregation, joins, and conditional logic

3

Customer Segments

New, Returning, and Loyal — enabling targeted campaigns

Data Cleaning & Preprocessing

Raw data is only as good as the care put into preparing it.

Missing Value Imputation

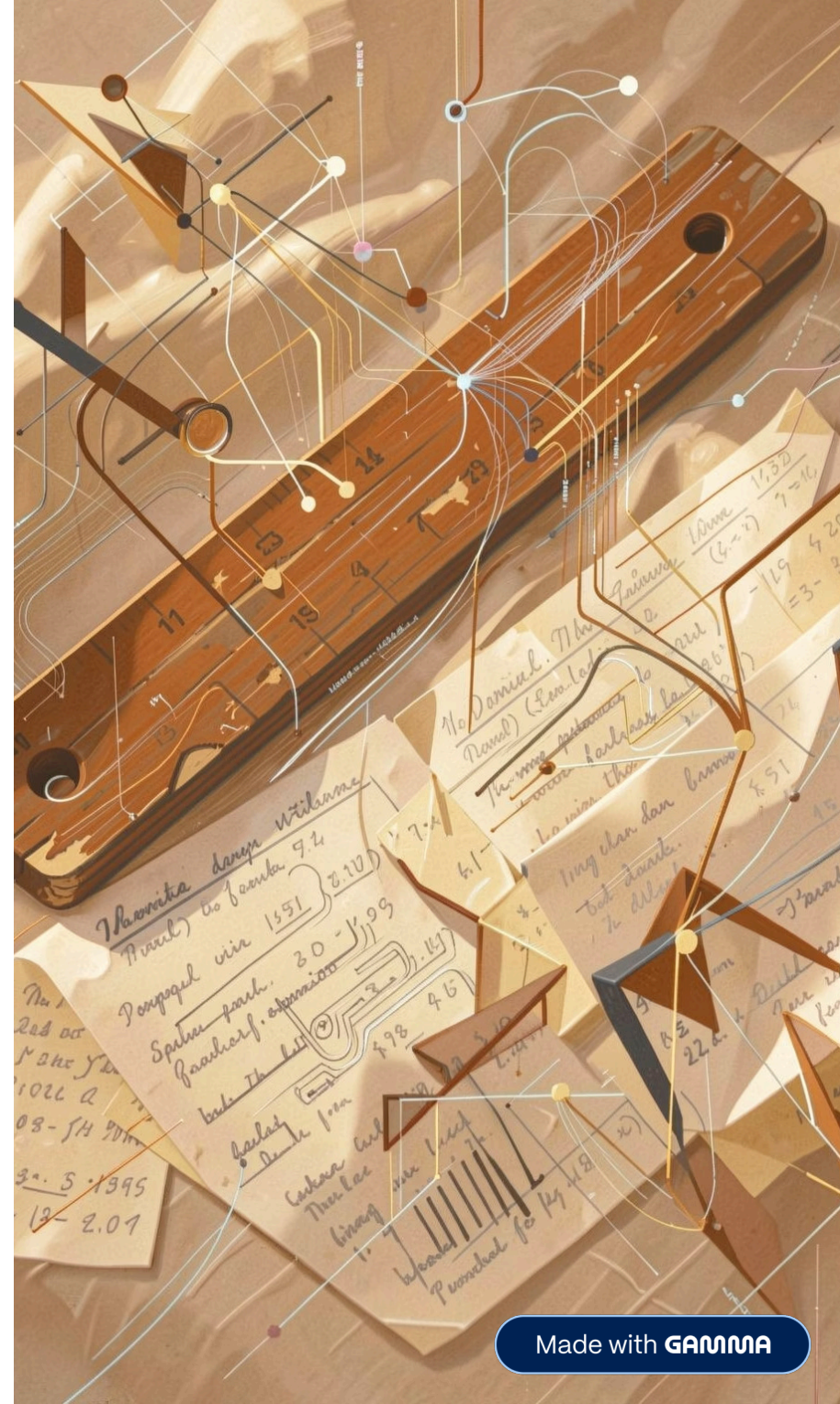
Addressed gaps in `Review Rating` by imputing category-level medians, preserving statistical integrity across product groups.

Feature Engineering

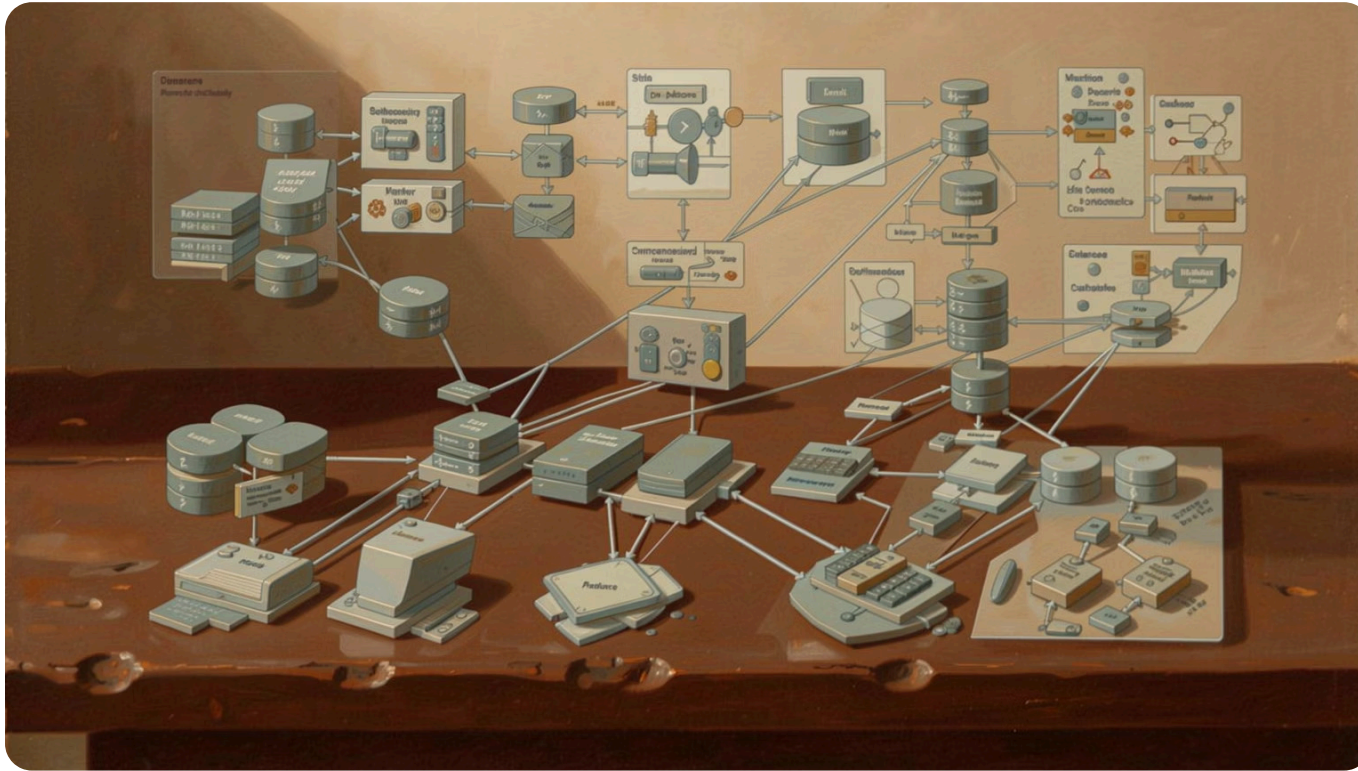
Created an `age_group` column via `pd.qcut` for demographic segmentation; mapped purchase frequencies to numerical day values for time-series readiness.

Standardization

Renamed inconsistent columns and transformed disparate datasets into a unified, analysis-ready schema.



Data Modeling & Database Management



SQLite Database Design

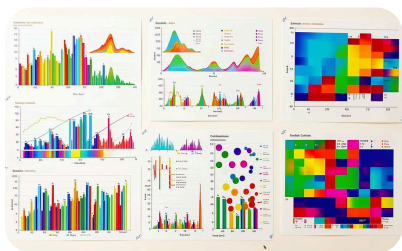
Designed and implemented a structured SQLite database to store cleaned customer data. Used `pandas.to_sql()` for efficient, repeatable data loading and replacement — keeping the pipeline clean and auditable.

Advanced SQL Querying

Executed complex queries using **CTEs**, **aggregation functions**, and **conditional logic** to extract customer segmentation insights and evaluate product performance at scale.

Exploratory Data Analysis & Visualization

Used **Matplotlib** and **Seaborn** to transform raw distributions into clear, decision-ready visuals.



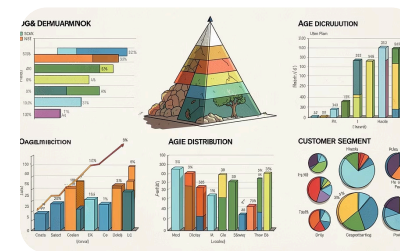
Distribution Analysis

Visualized item categories, age groups, and purchase amounts to surface underlying spending patterns.



Category Performance

Identified top-performing product categories by average review rating and average purchase amount.



Demographic Segmentation

Broke down spending behavior across age groups to surface actionable differences by life stage.

Business Insights & Impact

Subscription Impact

Revealed measurable differences in spending behavior between subscribers and non-subscribers, linking loyalty programs to revenue outcomes.

Customer Segmentation

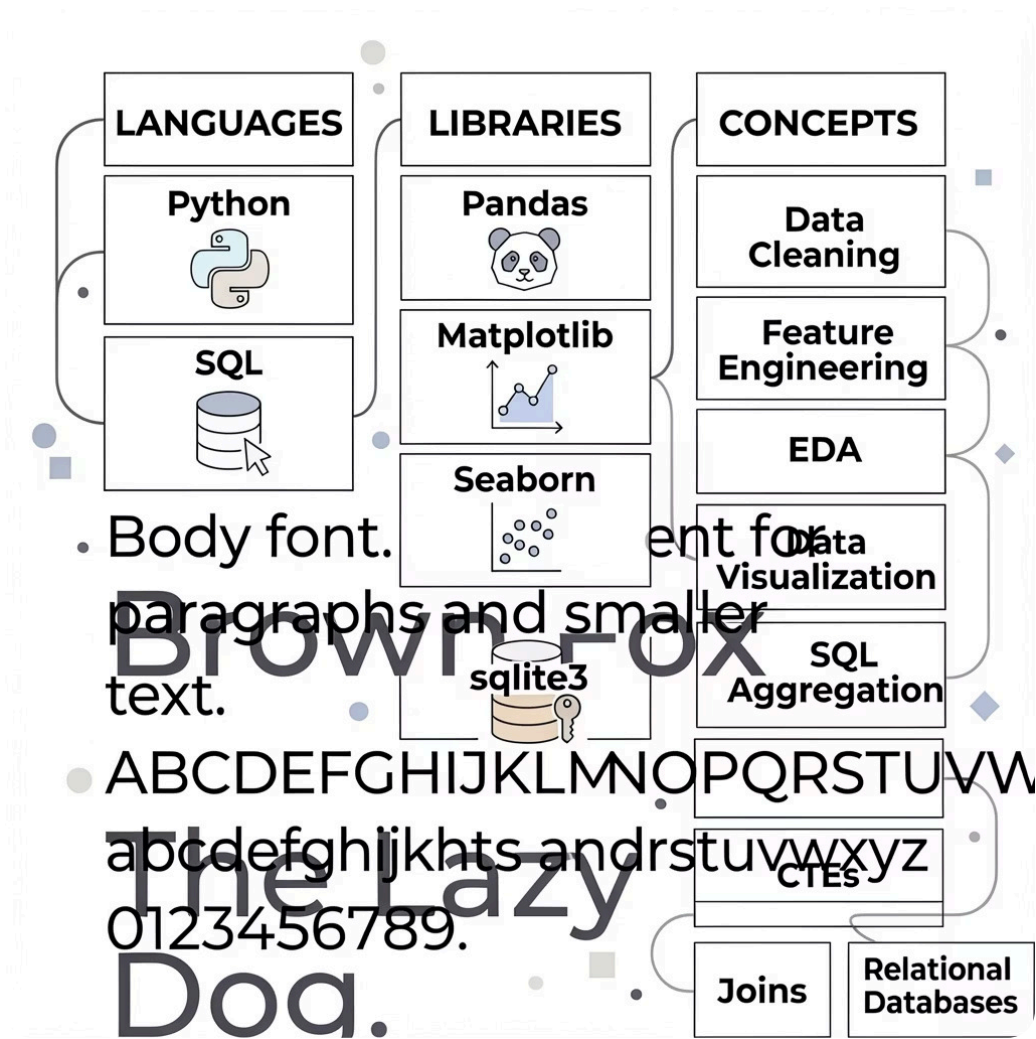
Classified customers as **New, Returning, or Loyal** based on purchase history — enabling targeted, lifecycle-driven marketing campaigns.

Discount Strategy

Identified the top 5 products with the highest discount usage rates, directly informing pricing and promotional planning.



Tools & Technologies



A Full-Stack Data Skillset

This project demonstrates proficiency across the complete analytics pipeline — from raw data ingestion and engineering through to relational database design and stakeholder-ready visualizations.

Every tool was chosen to reflect real-world production workflows used by data and analytics teams in retail and e-commerce.

The End-to-End Workflow



Each phase builds directly on the last — ensuring clean inputs, structured storage, rigorous analysis, and clear output for decision-makers.

Ready to Drive Data-Informed Decisions

This project is a repeatable template for how I approach analytics: start with data quality, build reliable infrastructure, ask the right business questions, and communicate findings with clarity.

"Cleaned, modeled, queried, and visualized — from raw CSV to retail insight."

